

Charotar University of Science and Technology

Sixth semester of B. Tech (IT) Examination

June 2022

IT348 Cryptography and Network Security

Date: 02/07/2022, Saturday

Time: 10:00 am to 1:00 pm

Maximum Marks: 70

Instructions:

- (i) Attempt *all* the questions.
- (ii) Figures to the right indicate *full* marks.
- (iii) Make suitable assumptions and draw neat figures wherever if required.

Section 1

Q-1 (a) Define the following [2]

- I. Network Security
- II. Interruption
- III. Interception
- IV. Modification

(b) Use the playfair cipher to encipher the message "The algorithm is very strong". Use the key "respective". [2]

(c) List and explain various approaches to evaluate security of a cryptosystem. [3]

Q-2 (a) Explain the working of Cipher Block Chaining. Write advantages and disadvantages of it. [5]

(b) Rajesh stole the encrypted message and later he finds the original message for the stolen text. State the attack and assess the key if the technique used for encryption is Hill Cipher. [5]

$$P.T. = \begin{pmatrix} 5 & 7 & 10 \\ 13 & 17 & 7 \\ 0 & 5 & 4 \end{pmatrix} \quad C.T. = \begin{pmatrix} 3 & 6 & 0 \\ 14 & 16 & 9 \\ 3 & 17 & 11 \end{pmatrix}$$

(c) Find the Greatest Common Divisor of 2740 and 1760 using Euclidean Algorithm. [4]

OR

Q-2 (a) Explain single round of DES with figure. [4]

(b) Explain in brief. [5]

1. Chosen-message Attack
2. Known-message Attack
3. Key-only Attack

(c) Encrypt the message using Playfair cipher "The house is being sold tonight" with the key "MONARCHY". [5]

Q-3 (a) List and explain cryptographic hash function criteria. [4]

(b) Explain MDC and MAC in detail. [5]

(c) Write a short note on Broadcast Security and Multicast-Rekeying. [5]

OR

Q-3 (a) What is message and message digest? Discuss the security of MAC in detail. [4]

- (b) What is Kerberos? List down its servers. Define the duties of each server [5]
 (c) What is PKI? Discuss the need and future of PKI. [5]

Section 2

- Q-4 (a) Solve the following by applying Chinese Remainder Theorem. [3]
 $a_1=2, a_2=3, a_3=3$ and $m_1=3, m_2=5, m_3=7$.
 (b) Discuss RSA Algorithm with suitable example [7]
 Q-5 (a) What is trapdoor function? What is the strength of trapdoor function in cryptography? [3]
 (b) Explain the invertible and non-invertible components used in Feistel cipher. [6]
 (c) Calculate the padding bit (SHA512) require for the messages having below-mentioned length: [3]
 1. 1
 2. 896
 3. 897

OR

- Q-5 (a) Differentiate active and passive attack. [3]
 (b) Write a short note on any TWO of the following [6]
 1. Master Key Generation in SSL 2. Trust calculation in Pretty Good Privacy
 3. Secure MIME
 (c) Draw and explain model for network security. [3]
 Q-6 (a) Find the multiplicative inverse of 23 in Z_{100} . [3]
 (b) Given $p = 31, q = 23, e = 223$ and m (plain text) = 439. Demonstrate the working of RSA algorithm using given values. [7]
 (c) What is cryptanalysis? What are the different ways to do cryptanalysis? [3]

OR

- Q-6 (a) What is the purpose of X.509 standard? Explain the signature field of X.509 certificate format. [3]
 (b) Explain Pretty Good Privacy and Symmetric Key Distribution. [7]
 (c) Explain Handshake protocol in SSL. [3]
